THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 17	The common difference is 3, so add 3 to the last term.
2	**(b)** 15	The 8th odd number is $2 \times 8 - 1$.
3	**(b)** 14	The 7th even number is 2×7.
4	**(d)** 36	$6 \times 6 = 36$.
5	**(c)** 10	$\frac{4 \times 5}{2} = 10$.
6	**(c)** 8	$2 \times 2 \times 2 = 8$.
7	**(b)** 16	The sum of the first n odd numbers is $n \times n$.
8	**(b)** 25	The common difference is 4, so add 4 to the last term.
9	**(b)** 23	The 12th odd number is $2 \times 12 - 1$.
10	**(b)** 22	The 11th even number is 2×11.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Common difference $= 5$ (1); Next three: $24, 29, 34$ (1).	1+1
12	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25$ (1).	1+1
13	Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15$ (1).	1+1
14	Common difference $= 3$ (1); Next three: $21, 24, 27$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Common difference $= 2$ (1); Term 8 $=$ first term $+7\times$ difference $= 3 + 7 \times 2$ (1); 17 (1).	1+1+1
16	$1 + 3 + 5 + 7 + 9 = 25$ (1); $25 = 5 \times 5$ is a square number (2).	1+2
17	$1 + 2 + \cdots + 9$ (1); $= 45$ (2).	1+2
	OR: Common difference $= 3$ (1); Term 7 $=$ first term $+6\times$ difference $= 5 + 6 \times 3$ (1); 23 (1).	1+1+1
18	$1 + 3 + 5 + 7 + 9 + 11 = 36$ (1); $36 = 6 \times 6$ is a square number (2).	1+2
	OR: $1 + 2 + \cdots + 10$ (1); $= 55$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First 6 terms: $2, 5, 8, 11, 14, 17$ (1); Term 7 is 20 and term 8 is 23 (2); Yes, 23 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 = 16$ (1); $16 = 4 \times 4$ (2); Sum of first 5 odd numbers $= 5 \times 5 = 25$ (2).	1+2+2
20	First 6 terms: $4, 9, 14, 19, 24, 29$ (1); Term 7 is 34 and term 8 is 39 (2); Yes, 39 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 + 9 = 25$ (1); $25 = 5 \times 5$ (2); Sum of first 6 odd numbers $= 6 \times 6 = 36$ (2).	1+2+2

Mistakes to avoid:

- Thinking the sum of the first n natural numbers is $n \times n$; that is the sum of the first n **odd** numbers.
- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 81	$9 \times 9 = 81$.
2	**(a)** 15	$\frac{5\times6}{2} = 15$.
3	**(c)** 27	$3 \times 3 \times 3 = 27$.
4	**(d)** 36	The sum of the first n odd numbers is $n \times n$.
5	**(b)** 17	The common difference is 2, so add 2 to the last term.
6	**(a)** 29	The 15th odd number is $2 \times 15 - 1$.
7	**(b)** 28	The 14th even number is 2×14.
8	**(c)** 121	$11 \times 11 = 121$.
9	**(a)** 21	$\frac{6\times7}{2} = 21$.
10	**(d)** 64	$4 \times 4 \times 4 = 64$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25, 36$ (1).	1+1
12	Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15, 21$ (1).	1+1
13	Common difference $= 7$ (1); Next three: $34, 41, 48$ (1).	1+1
14	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25, 36, 49$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Common difference $= 5$ (1); Term 9 $=$ first term $+8\times$ difference $= 2 + 8 \times 5$ (1); 42 (1).	1+1+1
16	$1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$ (1); $49 = 7 \times 7$ is a square number (2).	1+2
17	$1 + 2 + \cdots + 11$ (1); $= 66$ (2).	1+2
	OR: Common difference $= 4$ (1); Term 10 $=$ first term $+9\times$ difference $= 4 + 9 \times 4$ (1); 40 (1).	1+1+1
18	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 = 81$ (1); $81 = 9 \times 9$ is a square number (2).	1+2
	OR: $1 + 2 + \cdots + 12$ (1); $= 78$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First 6 terms: $1, 8, 15, 22, 29, 36$ (1); Term 7 is 43 and term 8 is 50 (2); Yes, 50 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 + 9 + 11 = 36$ (1); $36 = 6 \times 6$ (2); Sum of first 7 odd numbers $= 7 \times 7 = 49$ (2).	1+2+2
20	First 6 terms: $6, 10, 14, 18, 22, 26$ (1); Term 7 is 30 and term 8 is 34 (2); Yes, 34 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 = 49$ (1); $49 = 7 \times 7$ (2); Sum of first 8 odd numbers $= 8 \times 8 = 64$ (2).	1+2+2

Mistakes to avoid:

- Thinking the sum of the first n natural numbers is $n \times n$; that is the sum of the first n **odd** numbers.
- Guessing a rule from only two terms. Always check the rule on at least three terms.
- Mixing up the nth odd number $2n - 1$ with the nth even number $2n$.
- Treating $2, 4, 8, 16$ as having a common difference; the gaps are $2, 4, 8$, so no constant is added.
- In the Virahanka sequence, adding the wrong two terms (the rule uses the *two terms just before*).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 1: **Patterns in Mathematics** | Class 6 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** 64	The sum of the first n odd numbers is $n \times n$.
2	**(b)** 35	The common difference is 5, so add 5 to the last term.
3	**(b)** 39	The 20th odd number is $2 \times 20 - 1$.
4	**(b)** 36	The 18th even number is 2×18.
5	**(a)** 169	$13 \times 13 = 169$.
6	**(c)** 28	$\frac{7 \times 8}{2} = 28$.
7	**(c)** 125	$5 \times 5 \times 5 = 125$.
8	**(c)** 81	The sum of the first n odd numbers is $n \times n$.
9	**(d)** 33	The common difference is 6, so add 6 to the last term.
10	**(a)** 49	The 25th odd number is $2 \times 25 - 1$.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15, 21, 28$ (1).	1+1
12	Common difference $= 4$ (1); Next three: $31, 35, 39$ (1).	1+1
13	Squares are $1 \times 1, 2 \times 2, \ldots$ (1); $1, 4, 9, 16, 25, 36, 49, 64$ (1).	1+1
14	Add 1, then 2, then 3, ... one at a time (1); $1, 3, 6, 10, 15, 21, 28, 36$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Common difference $= 3$ (1); Term 8 $=$ first term $+7\times$ difference $= 6 + 7 \times 3$ (1); 27 (1).	1+1+1
16	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 = 100$ (1); $100 = 10 \times 10$ is a square number (2).	1+2
17	$1 + 2 + \cdots + 13$ (1); $= 91$ (2).	1+2
	OR: Common difference $= 6$ (1); Term 7 $=$ first term $+6\times$ difference $= 1 + 6 \times 6$ (1); 37 (1).	1+1+1
18	$1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 + 19 + 21 + 23 = 144$ (1); $144 = 12 \times 12$ is a square number (2).	1+2
	OR: $1 + 2 + \cdots + 14$ (1); $= 105$ (2).	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	First 6 terms: $3, 11, 19, 27, 35, 43$ (1); Term 7 is 51 and term 8 is 59 (2); Yes, 59 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 = 64$ (1); $64 = 8 \times 8$ (2); Sum of first 9 odd numbers $= 9 \times 9 = 81$ (2).	1+2+2
20	First 6 terms: $5, 11, 17, 23, 29, 35$ (1); Term 7 is 41 and term 8 is 47 (2); Yes, 47 is the 8th term (2).	1+2+2
	OR: $1 + 3 + 5 + 7 + 9 + 11 + 13 + 15 + 17 = 81$ (1); $81 = 9 \times 9$ (2); Sum of first 10 odd numbers $= 10 \times 10 = 100$ (2).	1+2+2

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